trait impl PSRN for TulapPSRN

Yu-Ju Ku, Jordan Awan, Aishwarya Ramasethu, Michael Shoemate

July 15, 2024

This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `TulapPSRN`.

`edge` accepts parameter `self`, containing the state of the Tulap sampler and `R` specifying the rounding mode.

This implementation is susceptible to floating-point vulnerabilities.

Warning 1 (Code is not constant-time)**.** The implementation of `edge` uses procedures that are vulnerable to timing attacks.

PR History

- Pull Request #1126

1 Hoare Triple

Preconditions

- Variable `self` is of type `TulapPSRN`.
- Generic `R` denotes the rounding mode, one of "up" or "down".

Pseudocode

```


class TulapPSRN(object):
    def __init__(self, shift, epsilon, delta) -> None:
        self.shift = shift
        self.epsilon = epsilon
        self.delta = delta
        self.uniform = UniformPSRN()
        self.precision = 50

    def q_cnd(self, u, c, R):  # CND quantile function for f
        if u < c:
            return self.q_cnd(1 - self.f(u, R), self.f(u, R), c) - 1
        elif c <= u <= 1 - c:  # the linear function
            return (u - 1 / 2) / (1 - 2 * c)
        else:
```

```
            return self.q_cnd(self.f(1 - u, R), self.f(u, R), c) + 1

    def f(self, u, _R):
        epsilon, delta = self.epsilon, self.delta
        return max(0, 1 - delta - exp(epsilon) * u, exp(-epsilon) * (1 - delta - u))

    def edge(self, R):
        epsilon, delta = self.epsilon, self.delta
        unif = self.uniform.edge(R)
        c = (1 - delta) / (1 + exp(epsilon))
        if c == 0.5:
            return None

        return self.q_cnd(unif, c, R)

    def refine(self):
        self.precision += 1
        self.uniform.refine()

    def refinements(self):
        return self.precision
```

Postcondition

`edge` returns an estimate of the true Tulap sample, a distribution with CDF defined in `make_tulap`. This mechanism is not implemented in a fashion that accurately samples from the Tulap distribution, so it may be subject to artifacts.

2 Proof

Proof. The cdf of $\text{Tulap}(0, b, q)$ is

$$F_N(x) = \begin{cases} 0 & F_{N_0}(x) < q/2 \\ \frac{F_{N_0}(x) - q/2}{1-q} & q/2 \leq F_{N_0}(x) \leq 1 - q/2 \\ 1 & F_{N_0}(x) > 1 - q/2. \end{cases}$$

By inspection, the fixed point of $f_{\epsilon,\delta}$ is $c = \frac{1-\delta}{1+e^{\epsilon}}$. It is easy to verify that $F_N(x) = c(1/2 - x) + (1-c)(x + 1/2)$ for $x \in (-1/2, 1/2)$.

The function then uses the inverse transform of a sample of a uniform RV to sample a Tulap RV centered at zero. The function then returns the value, shifted by `self.shift`.

□